Exploratory Data Analysis using Visualization

Assignment

PW SKILLS

Dataset: Titanic_Dataset

Problem Statement – Titanic Passenger Analysis

You are a data analyst studying passenger data from the Titanic ship. The dataset contains information about passenger class, age, gender, fare, family members, ticket details, and survival status.

The objective of this assignment is to perform Exploratory Data Analysis (EDA) using Matplotlib and Seaborn to understand passenger distribution, survival patterns, and relationships between different variables using data visualization.

You need to create appropriate visualizations, interpret the charts, and generate insights from the data.

Submission Requirements

Students must submit:

1. Google Colab Notebook
2. Visualizations with titles
3. Interpretation for each chart
4. Final insights
5. Clean and readable code

Assignment Tasks

Instructions

Students must:

1. Load the Titanic dataset.
2. Perform basic data exploration using info() and describe().
3. Handle missing values if necessary.
4. Create the required visualizations.
5. Add title and axis labels to each chart.
6. Write interpretation for each visualization.
7. Write final insights from the dataset.

Assignment Questions (Visualization Based)

Q1. Create a countplot to show the number of passengers by gender.

Q2. Create a bar chart to show the number of passengers in each passenger class using different colors.

Q3. Create a histogram to show the distribution of passenger age.

Q4. Create a boxplot to detect outliers in the fare column.

Q5. Create a scatterplot to

a. show the relationship between age and fare.

b. analyze the relationship between family size (SibSp + Parch) and fare.

Q6. Create a countplot to show survival count by gender.

Q7. Create a bar chart to show survival by passenger class.

Q8. Create a heatmap to show correlation between numerical variables.

Q9. Identify which passenger class paid the highest fare on average.

Q10. Write at least 5 insights from the dataset based on your visualizations.